

Homework — 1.4.2 Data Structures

Name: _____ Date: _ **Mark:** _____ / 20

OCR H446 — set after the 1.4.2 lesson. *SHOW ALL WORKING.*

Part A — This week's topic (~14 marks)

1. A **stack** is initially empty. The following operations are carried out in order. Show the stack after each operation, then give the values popped (in order) and the final contents. **[4]** (AO2)

`push(8)` , `push(3)` , `push(6)` , `pop()` , `pop()` , `push(1)`

Working space (show the stack after each operation):

Values popped (in order): _ **Final stack (top → bottom):** _

2. Explain why a **circular queue** is often preferred over a simple linear queue that uses front and rear pointers in a fixed array. Refer to how the rear pointer is updated. **[3]** (AO1)

Working space:

3. The following numbers are inserted, in this order, into an empty **binary search tree**: **40, 60, 25, 35, 10, 50**. Draw the resulting tree. **[3]** (AO2)

Working space:

4. Give the **in-order** traversal of the binary search tree you drew in Question 3, and state what is special about the sequence produced. [2] (AO2)

Working space:

5. A hash table of size **10** uses the hash function $\text{index} = \text{key} \bmod 10$. The keys **23, 57, 33** are inserted in that order. State the index each key hashes to, identify any **collision**, and name **one** method of resolving it. [2] (AO2)

Working space:

Part B — Retrieval: keep it fresh (~6 marks)

6. Explain why an operating system uses a **LIFO stack** to handle **nested interrupts**. (1.2) [2] (AO1)

Working space:

7. State the role of the **control bus** in a processor. (1.1) [1] (AO1)

Working space:

8. A relational database links two tables using a **foreign key**. State what a foreign key is, and explain what **referential integrity** ensures. (1.3) [3] (AO1)

Working space: